# WaveWise

# Final Report Document

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12/22/2023


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# Goal

The goal of the final report is to enable the client to continue the project without needing to contact  Generate.

### General Guidelines

This document is the primary reference for the client after their involvement with Generate is complete. The client should be able to understand the entire development process without additional explanation from Generate team members.

Emphasize the thought process behind decisions.

Place all illustrations in the document's body sections. And place all images, code, testing data, etc., in the appendices at the end of the document.

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# Executive Summary

### Purpose

Currently, Maine's capacity for sustainable sugar kelp production is hindered by numerous factors including: location and nutrient restrictions, limited and ineffective monitoring capabilities, and overexposure of crops to ultraviolet radiation. The client, WaveWise, approached Generate requesting a prototype of an aquaculture system capable of dynamic depth adjustment to achieve optimal kelp growth.

Throughout the course of the semester, the team worked on researching, ideating, designing, constructing, and testing a system that would meet the requirements outlined by the client. Ultimately, the team progressed with a design featuring a winch system which would adjust the depth of a kelp-bearing submersible. Depth control was facilitated by a stepper-motor-powered drivetrain that operated as a function of various aquaspheric variables. These variables were read in by an array of sensors affixed to the submersible and relayed to the winch system via a waterproof cable. The sensor readings were also written to a text file and saved locally on an SD card for future user analysis. The winch system was housed on a buoy which provided sufficient buoyancy force to support the weight of the entire system. Additionally, the buoy housed the infrastructure for a four-panel solar array, which transmitted power to be stored in a lithium ion battery. Power to the submersible was provided via the same waterproof cable through which the sensor data was transmitted.

The team learned and honed a variety of skills—most notably computer-aided design (CAD), designing for assembly (DFA), electromechanical integration, and waterproofing on the mechanical side, as well as PCB design, layout, and bringup, component spec'ing, and firmware development with regards to electrical and computer development.

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# Background Research

Aquaculture is currently dominated by traditional methods that do not incorporate dynamic growing platforms. While studies have been conducted on automated methods of seaweed monitoring cultivation, these focus on monitoring the harvesting process using automated underwater vehicles (AUVs) as opposed to dynamically adjusting the kelp depth in response to surrounding environmental conditions [1].

During the initial research and ideation phase, the leads worked together with the client to develop a list of deliverables to be achieved by the end of the semester. These deliverables were created with the goals of the client in mind as well as considering the abilities and limitations of Generate as a student-led product development studio with aggressive budgetary and timeline constraints. Ultimately, 11 product requirements were created at the start of the semester, and can be found below:

1. **This product shall be designed to integrate with pre-existing aquaculture infrastructure.**

   Integration with standard, currently-existing aquacultural infrastructure was an essential requirement to make a commercially viable consumer product. While the client requested a prototype as opposed to a finalized consumer product, it remained important to create a design that could be easily iterated upon with the ultimate goal of consumer viability.

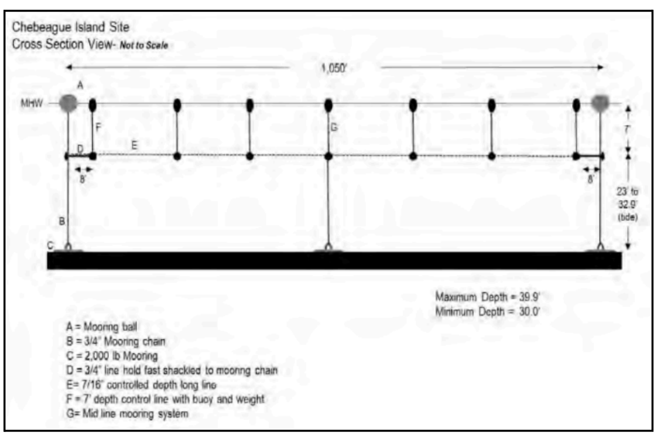



**Figure 1.** A cross-sectional view of a typical kelp farm found off the coast of Maine.

The information gathered regarding kelp farming was sourced from Flavin et al.'s "Kelp Farming Manual A Guide to the Processes, Techniques, and Equipment for Farming Kelp in New England Waters" [2], which proved invaluable in understanding current industry practices and conditions. From the details outlined in this guide, there were three major components that comprised a kelp farm (see left): moorings and ground tackle, long lines, and depth maintenance systems [2, p. 6]. As the client's requested product would serve the function of a depth maintenance system, this would need to be integrated with the remaining two subsystems—moorings and ground tackle and long lines.

The mooring and ground tackle systems—which "consist of 2,000-lb. high density concrete blocks" pictured in Figure 2 on the right—fix the farm to the ocean floor and prevent any drifting [2, p. 7]. Vertical lengths of "7/16-inch poly line" are tied to these moorings, with the other end affixed to "a 200-lb. (or greater) displacement buoy" [2, p. 7]. These buoys are then linked together via "long lines"—1,000 to 1,500 feet [2, p. 5] of conjoined 200-foot lengths of 7/16-inch poly line [2, p. 7]. Long lines run parallel to one another, typically spaced from 5 to 15 feet apart [2, p. 6].

**Figure 2.** Standard concrete kelp farm moorings.

2. **This product shall dynamically adjust the depth of an aquaculture platform.**

In a traditional kelp farming setup, "each 200-foot long line section" contains a "depth control dropper" shown below in Figure 3 [2, p. 7]. These depth control droppers feature a small buoy tethered to a 10-pound weight via a 7-foot length of pipe. Under this configuration, "the buoy holds the line at the desired depth when the kelp is small," until "the kelp grows larger [and] the stipes fill with gas" [2, p. 7]. Once this occurs, the kelp becomes buoyant and the weight holds the long line at the desired depth.

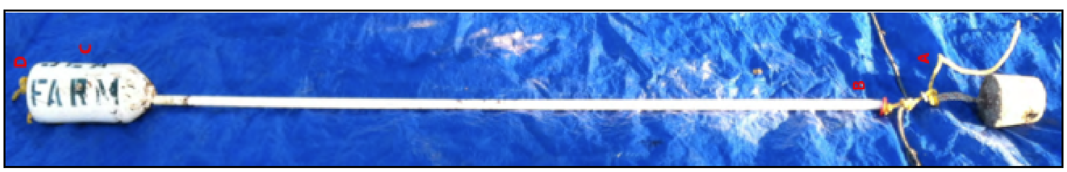



**Figure 3.** A standard dropper in a kelp depth maintenance system.

The issue with this type of depth maintenance system is that it requires that the kelp reside at a constant depth. To maximize kelp growth requires the optimization of a variety of environmental conditions, including water temperature, salinity, pH, light, and nitrogen concentration [2, pp. 38, 85]. Optimal sugar kelp growth conditions can be found in Table 1 to the right.

**Table 1.** Optimal kelp growth conditions with respect to a variety of variables [2, pp. 38, 85] [3].

| Parameter | | Low Range | High Range |
|---|---|---|---|
| Temperature | | 41 ℉ / 5 ℃ | 59 ℉ / 15 ℃ |
| Salinity | | 28 ppt | 34 ppt |
| pH | | 7.0 | 9.0 |
| Light | 1 - 14 days | 20 µmol/m$^2$ · s / 1080 lux | |
| | 15 - 28 days | 55 µmol/m$^2$ · s / 2970 lux | |
| | 29+ days | 100 µmol/m$^2$ · s / 5400 lux | |
| Nitrate | | 1 µmol/L | 4 µmol/L |

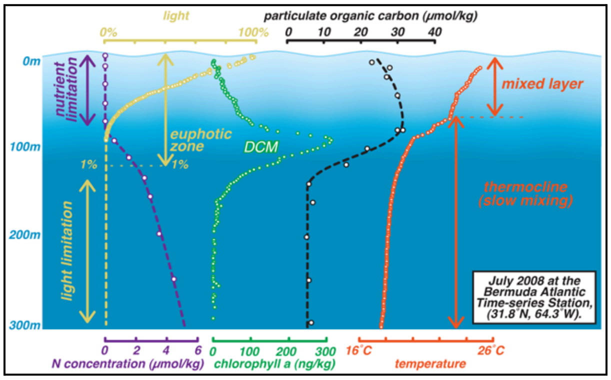



**Figure 4.** Various oceanic environmental conditions as functions of depth [4].

Generally, these conditions are not constant throughout the ocean, but change as functions of depth. This can be observed to the left in Figure 4, where numerous variables influential towards kelp growth are plotted with respect to depth. While the figure illustrates data collected near Bermuda—which observes vastly different conditions compared to those off the coast of Maine—it does an excellent job showcasing how environmental conditions vary with respect to depth. It is also worth noting that these variables are functions of numerous long and short term changes to environmental oceanic conditions, ranging from time of day to oceanic heat waves to climate change [5].

Under the current depth maintenance systems traditionally utilized in Maine's kelp farming, not only is the kelp fixed at a constant depth, but this depth is typically only a few meters below surface level [2, p. 8]. While this setup does ensure that the kelp receives sunlight required to facilitate growth, it prevents the kelp from receiving the abundance of nutrients available at greater depths. This can be resolved via a dynamically adjusting system, which allows for the kelp to be depth-cycled. A 2021 study which compared the growth and health of depth-cycled kelp to stationary kelp found that "kelp grown under depth-cycling conditions had an average growth rate of 5% per day and produced four times more biomass... than individuals grown in a kelp bed without depth-cycling" [6].

3. **This product shall include various sensors (ie. pH, salinity, and nitrate concentration) which will monitor environmental conditions relating to optimal kelp growth.**

As can be observed in Table 1 above on page 5, there are very explicit ranges of temperature, salinity, pH, light, and nitrate concentration which are optimal for sugar kelp growth. An explicit list of sensors was not required, but the sensors listed above proved likely candidates of sensors that could be introduced in the system.

4. **This product shall be designed to be self powered.**

   Due to the system's location in the ocean, it is unreasonable to expect to be able to connect the system to an existing power grid. Therefore, it is imperative that the system be able to generate sufficient power to remain indefinitely operable on its own. Neither the type of power generation nor the amount of power required were explicitly outlined.

5. **This product shall function up to a depth of at least 60 feet (18 meters) underwater.**

   While the benefits of depth cycling typically start at depths greater than 60 feet, this was deemed a significant enough depth to demonstrate a proof-of-concept system. The depth of 60 feet was chosen as it was deemed a healthy compromise between the ultimate product vision and what could feasibly be accomplished given the team's constraints.

6. **This product shall gather and collect data relevant to kelp growth optimization.**

   Data collection was an integral element to the product. While data collection was an explicit requirement, the exact format and contents of the data were not specified due to the uncertainty regarding the type of sensors which would be integrated in the system.

7. **This product shall have a means of relaying collected data to a user.**

   It was essential that the system not only collect data, but possessed a means of relaying this data to a user so that the data could be subjected to external analysis. The means of data transmission was not specified, leaving both wireless communication and data transmission via a physical storage device as plausible paths forward.

8. **This product shall be designed to be manufactured from corrosion resistant materials.**

   As the product was designed for use in the ocean—a highly corrosive environment—it was deemed essential that the system be designed to be manufactured from corrosion resistant materials. However, due to budgetary constraints and the high cost of corrosion resistant materials, the prototype delivered at the end of the semester was not required to be made from corrosion resistant materials, so long as parts susceptible to corrosion could be easily replaced with corrosion resistant replacements in future design iterations.

9. **This product shall be designed with temperature regulation in mind.**

With the sugar kelp season typically starting in late October to November with seeding and concluding in March to May with harvesting [2, p. 4], the product would be required to operate under harsh conditions. Winter months off the coastline of Maine typically observe average air temperatures below 0℃, although temperatures below -20℃ are not uncommon [7].

**Table 2.** Climatic conditions in Eastport, Maine—one of the northernmost locations in Maine where sugar kelp is commonly harvested. Note the 'mean daily minimum' temperatures in the months of November through May [7].

| Climate data for Eastport, Maine | | | | | | | | | | | | | [hide] |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Month | Jan | Feb | Mar | Apr | May | Jun | Jul | Aug | Sep | Oct | Nov | Dec | Year |
| Record high °F (°C) | 58 (14) | 54 (12) | 76 (24) | 81 (27) | 90 (32) | 92 (33) | 93 (34) | 93 (34) | 92 (33) | 83 (28) | 71 (22) | 60 (16) | 93 (34) |
| Mean daily maximum °F (°C) | 29.8 (−1.2) | 32.1 (0.1) | 39.0 (3.9) | 49.9 (9.9) | 59.9 (15.5) | 68.5 (20.3) | 74.1 (23.4) | 73.8 (23.2) | 66.6 (19.2) | 55.9 (13.3) | 46.0 (7.8) | 35.6 (2.0) | 52.7 (11.5) |
| Daily mean °F (°C) | 21.7 (−5.7) | 24.3 (−4.3) | 31.4 (−0.3) | 41.6 (5.3) | 50.6 (10.3) | 58.4 (14.7) | 64.0 (17.8) | 64.1 (17.8) | 57.8 (14.3) | 48.2 (9.0) | 39.3 (4.1) | 28.2 (−2.1) | 44.2 (6.8) |
| Mean daily minimum °F (°C) | 13.7 (−10.2) | 16.5 (−8.6) | 23.7 (−4.6) | 33.5 (0.8) | 41.3 (5.2) | 48.3 (9.1) | 53.8 (12.1) | 54.4 (12.4) | 48.9 (9.4) | 40.5 (4.7) | 32.6 (0.3) | 20.8 (−6.2) | 35.8 (2.1) |
| Record low °F (°C) | −16 (−27) | −14 (−26) | 2 (−17) | 2 (−17) | 24 (−4) | 30 (−1) | 44 (7) | 42 (6) | 30 (−1) | 22 (−6) | 8 (−13) | −9 (−23) | −16 (−27) |
| Average precipitation inches (mm) | 3.8 (97) | 3.1 (79) | 4.1 (100) | 3.7 (94) | 3.8 (97) | 3.7 (94) | 3.0 (76) | 3.1 (79) | 4.1 (100) | 4.4 (110) | 4.9 (120) | 4.3 (110) | 45.8 (1,160) |
| Average precipitation days | 12.8 | 10.7 | 11.7 | 11.9 | 12.8 | 12.8 | 11.1 | 10.2 | 10.1 | 10.9 | 12.2 | 13.5 | 140.7 |
| Source: Weatherbase[17] | | | | | | | | | | | | | |

Temperatures this low often result in a variety of electrical and mechanical failures [8] [9], which could facilitate the need for some form of heating subsystem to be implemented. However, due to time and budgetary constraints, temperature regulation was determined to be a reach goal for the system. As a result, the team was to design the product with the electrical and mechanical capacity for future implementation of a temperature regulation subsystem whilst not necessarily including said subsystem.

10. **This product shall possess a modular and scalable design and form factor.**

Modularity and scalability were set as requirements with respect to the feasibility of integrating the product into existing infrastructure. This means that the product needed to be of a size and scale that would not deter kelp farmers from integrating the system into an existing farm.

11. **This product shall require minimal user input and interaction in order to set up and operate.**

Minimizing user input and interaction would ensure that the product would be easy for operators to understand and use.

# Systems Overview

WaveWise introduces a dynamic depth-adjustment system that allows kelp to be depth-cycled in accordance with real-time environmental feedback to achieve and maintain optimal growth conditions. WaveWise consists of three major subsystems—the support buoy, depth adjustment, and submersible—which are outlined below.

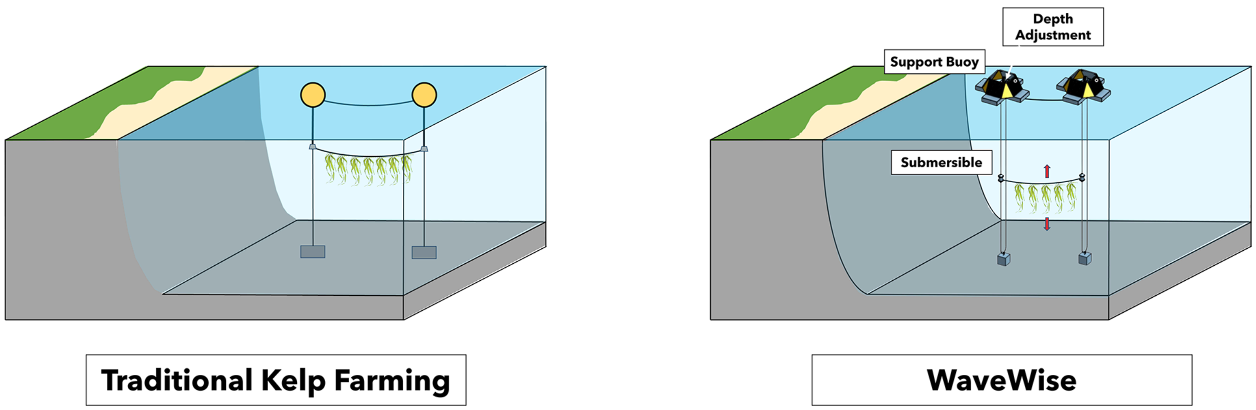



**Figure 5.** A diagram comparing a traditional kelp farm with a kelp farm utilizing WaveWise's dynamic depth adjustment system.

## Support Buoy

The support buoy is comprised of four 22" x 21" x 6" floatation blocks of poured polyurethane foam, which generate 600lbs worth of buoyancy force. These blocks were custom designed and cast in particle board molds to fit the specific geometric and floatation requirements of the system. 80/20 slotted aluminum framing was embedded within the foam-poured blocks, providing both structural support and a modular mounting system. A 12 volt solar panel was mounted to each of the foam blocks, using custom-printed brackets to affix to the 80/20 framing. Four foam blocks were then attached to one another using 80/20 compatible hardware and custom laser-cut acrylic paneling, with the solar panels connecting in series to facilitate a 250 Watt power output.

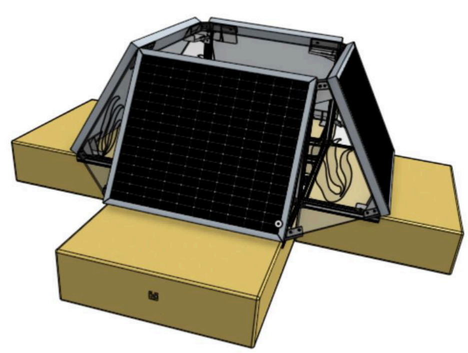


**Figure 6.** An isometric view of the support buoy CAD.

The output from the solar panels was used to charge a 50 volt rechargeable lithium-ion battery responsible for powering the entire electrical system. This 50 volt supply was bucked to a 5 volt rail for the majority of onboard electronics, and then shifted to a 3V3 volt rail for the data storage circuit.

These electronics are housed in IP68-rated enclosures, which are secured to a central acrylic panel supported by and secured to two opposite foam blocks. Additionally, this acrylic panel provides support for the depth adjustment system and its corresponding electrical and mechanical components.

## Depth Adjustment

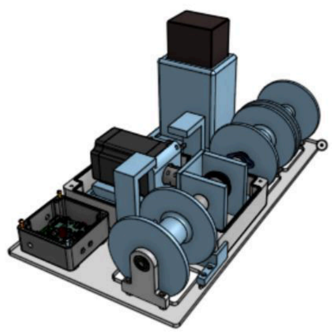


**Figure 7.** An isometric view of the depth adjustment system CAD.

The depth adjustment system controls the depth of the submersible in response to the sensor readings of surrounding environmental conditions. A winch system is utilized to simultaneously let in and out rope at the same rate, moving the submersible up or down whilst ensuring the same amount of rope is always submerged. The rope is spooled on two custom-printed winch barrels, which can let out rope to a maximum depth of 60 feet. The winch barrels are fixed to two contra-rotating coaxial shafts—or two coaxial shafts rotating in opposite directions—driven by a 8.5 Newton-meter stepper motor.

The motor shaft contains a beveled planet pinion which interfaces with two parallel beveled sun gears at a 90 degree angle. This causes the sun gears to rotate in opposite directions while increasing the torque capacity of the drivetrain via a 2:1 gear ratio. The stepper motor is directly powered from the battery and is capable of operating at 60 volts with a maximum draw of 6 amps, yielding a maximum power output of 360 watts.

Motor control is determined by onboard logic that operates as a function of sensor readings communicated via UART from the submersible to the depth adjustment system. This data is also saved on a 16GB microSD card contained on the buoy. The current SD card circuit is designed for SPI communication.

## Submersible

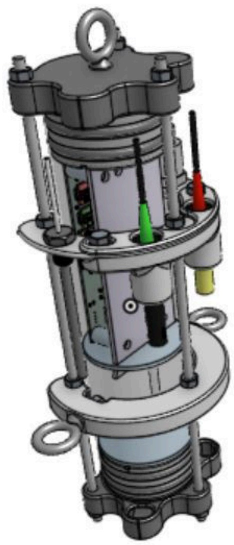
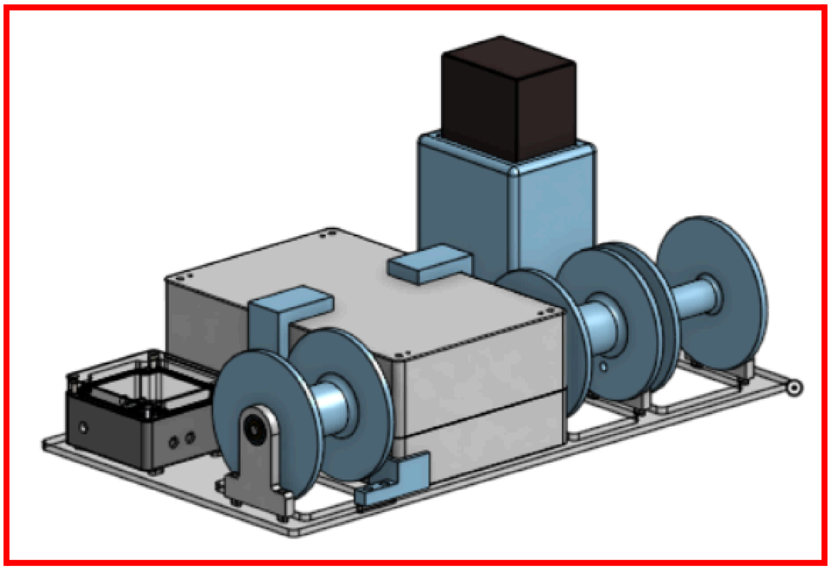


Figure 8. An isometric view of the submersible system CAD.

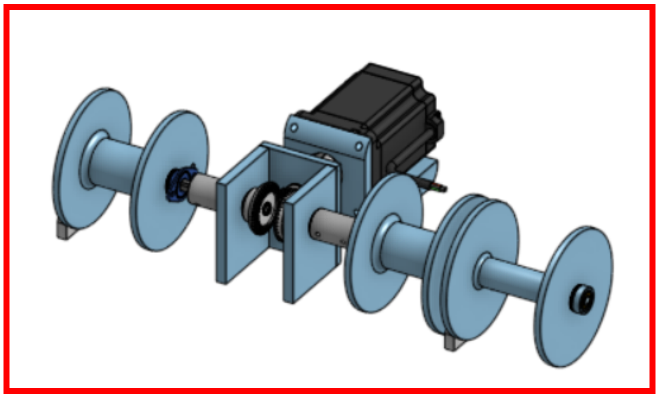
The submersible reads in surrounding environmental conditions using an array of sensors before relaying the information to the depth adjustment system via UART transmitted through a waterproof cable. The submersible is comprised of a clear acrylic pipe three inches in diameter by a foot long, enclosed by two custom printed, double o-ring sealed caps. Power and sensor wiring enters and exits through two designated, epoxy-sealed access points on the caps, creating a water-tight seal. The caps also contain eyebolts, which are used to attach the submersible to the depth adjustment system ropes via carabiners. Three threaded rods run the length of the submersible, constraining the caps in the vertical direction and providing a mounting fixture for the external sensors as well as a pair of outward-facing eyebolts. These eyebolts are attached to the kelp long lines via carabiners, facilitating the transfer of vertical translation from the submersible to the long lines.

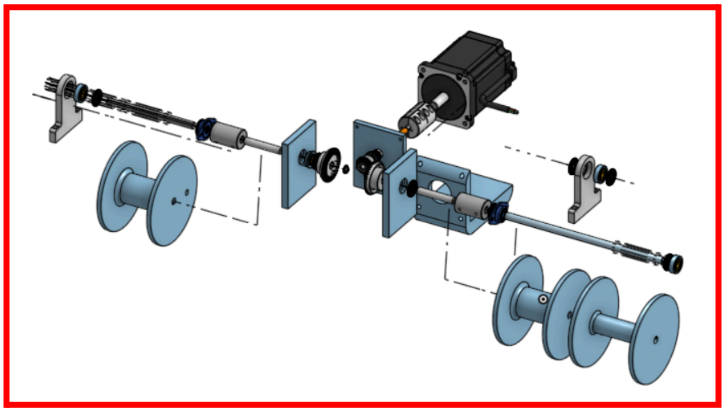
The submersible houses five different sensors which monitor the surrounding environmental conditions of the ocean relevant for kelp growth. These sensors include: pH, conductivity, temperature, pressure, and ultraviolet light. The ultraviolet light sensor is the only sensor contained within the submersible, and is mounted directly to the submersible's PCB. The remaining sensors are externally mounted and are fully waterproof, capable of prolonged exposure to the saltwater environment. Information is received from the sensors every 10 seconds, whereby the raw data is converted into relevant metrics and sent via a software-imitated UART bus to the buoy. The submersible PCB receives power from the buoy via waterproof cabling.

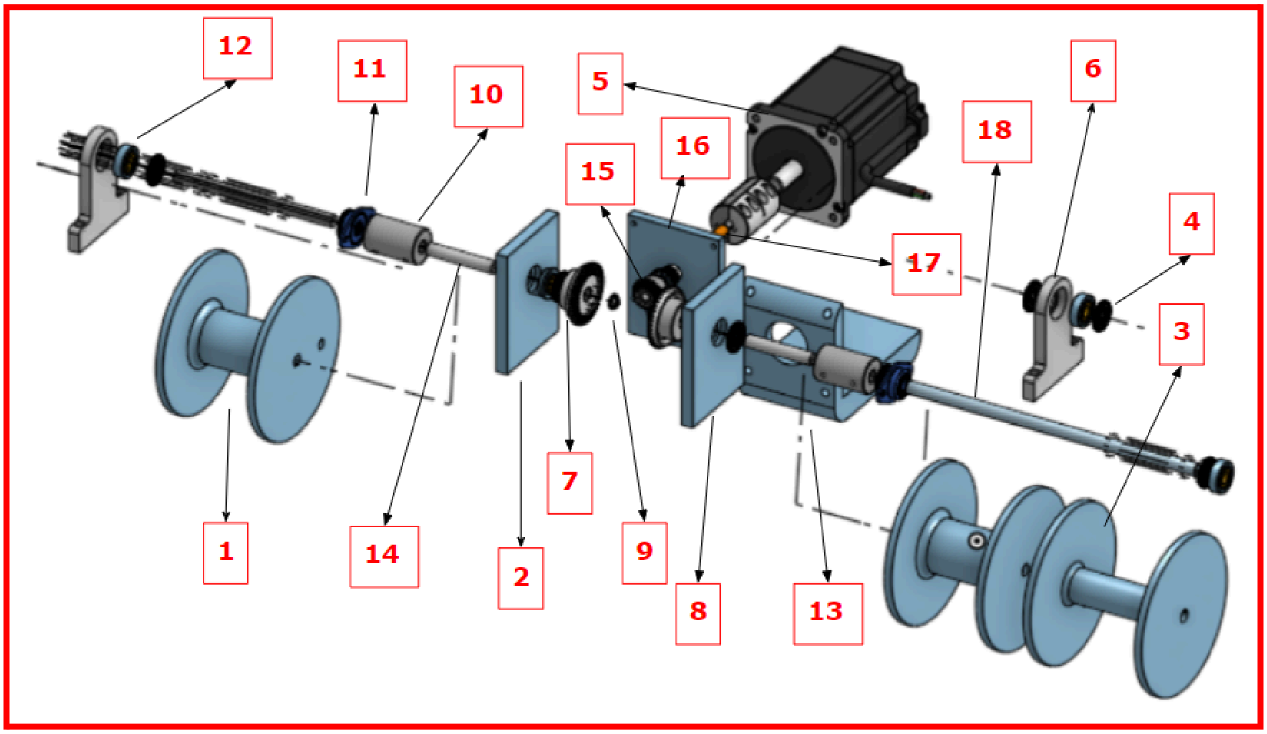
# Depth Adjustment System



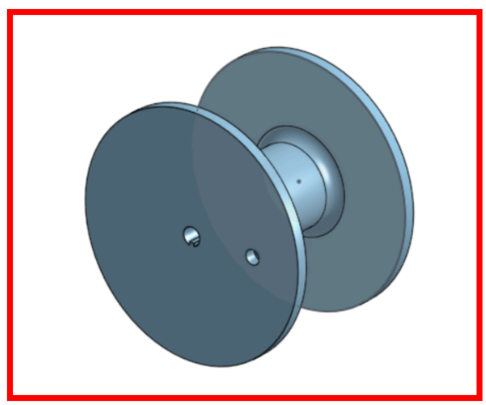
**Winch Subassembly**



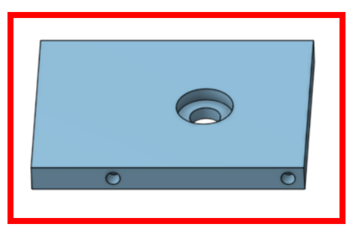





1. Winch Barrel. Rope - *Used for holding up to 100 ft of rope to hoist submersible and kelp lines in water; contains an extruded divet key to mesh to interface with keyed shaft*
    - Material: PLA
    - Flange Diameter: 5.25 in
    - Barrel Diameter: 1.75 in
    - Traverse Length: 3 in
    - Rope Diameter: ⅛ in
    - Hole Diameter for Keyed Shaft: 11.5 mm

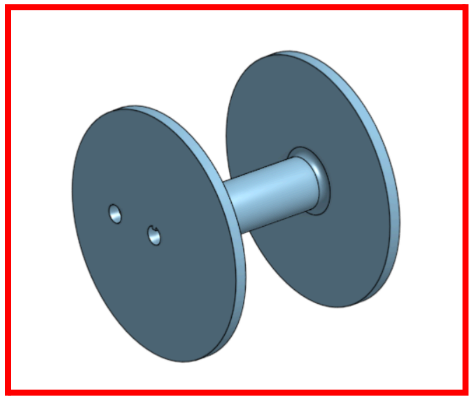


2. Right Side Bracket - *Used for gear alignment of driven gear in gear train and features two heat set inserts to be connected with the Top Bracket*
    - Material: PLA
    - Hole Diameter for Heat Set Insert: 5.5 mm
    - Length: 80 mm



- Height: 106.3 mm
- Thickness: 10 mm:

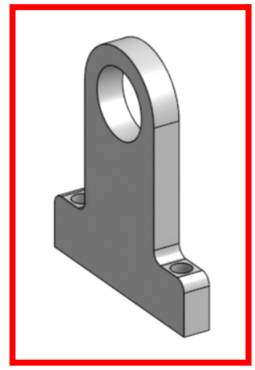
3. Winch Barrel, Cable - *Used for holding up to 60 ft of cable to powering sensors and electronics from the buoy to the submersible; contains an extruded divet key to mesh to interface with keyed shaft*
    - Material: PLA
    - Flange Diameter: 5.25 in
    - Barrel Diameter: 1.1 in
    - Traverse Length: 3.7 in
    - Cable Diameter: ¼ in
    - Hole Diameter for Keyed Shaft: 11.5 mm



4. Thrust Bearing, 10mm - *Used to decrease friction between two points of contact, specifically a fixed body and a rotating body*
    - Inner Hole Diameter: 10.03 mm
    - Outer Diameter: 24 mm
    - Thickness: 2 mm
    - Roller Material: Steel
    - Cage Material: Steel
    - Thrust Load Capacity:
        - Dynamic: 2050 lbs
        - Static: 5700 lbs
    - Max Speed: 17,000 rpm
    - Temperature Range: 0-240 deg Fahrenheit

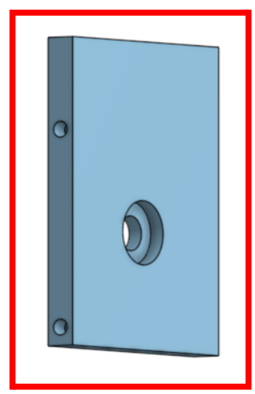
5. NEMA 34 Stepper Motor - *Responsible for driving gear train holding load of the submersible and kelp line*
    - Holding Torque: 8.5 Nm
    - Step Angle: 1.8 deg/step
    - Rated Current/Phase: 6 A
    - Voltage: 36-60 V
    - Shaft Diameter: 14 mm
    - Shaft Length: 32 mm
    - Weight: 5.5 kg
    - Keyway Shaft Length: 25 mm

6. Shaft End Supports - Responsible for clamping to acrylic panel with 2 countersunk M4 screw holes and allowing shaft to enter through a sealed bearing placed within the support
    - Material: PLA
    - Hole Diameter: 26.15 mm
    - Thickness: 3/8in



7. Bevel Gear, 40 tooth - *Used for gear alignment of driven gear of gear train and features two heat set inserts to be connected with the Top Bracket; an M4 was tapped into the hub of the gear to add a set screw*
    - Face Width: 8 mm
    - Module: 1.25
    - Pressure Angle: 20 deg
    - Gear Pitch Diameter: 50 mm
    - Overall Width: 19 mm
    - Shaft Diameter: 10 mm
    - Hub Diameter: 32 mm
    - Hub Width:  10 mm
    - Material: Black Oxide 1045 Carbon Steel

8. Left Side Bracket- *Used for gear alignment of driven gear in gear train and features two heat set inserts to be connected with the Top Bracket*
    - Material: PLA
    - Hole Diameter for Heat Set Insert: 5.5 mm
    - Length: 80 mm
    - Height: 106.3 mm
    - Thickness: 10 mm:



9. Retaining Ring, 10mm (2x) - *Constrain bodies from sliding that may occur in the shaft*
    - Material: DIN 1.4122 Stainless Steel
    - Outer Diameter: 10 mm
    - Thrust Load Capacity: 890 lb

**10.** Shaft Coupler (2x) - *connector between rounded shaft attached to driven gears to keyed shaft for the winch drums*

- Material: 1215 Carbon Steel Bar
- Bore Dimension: 10 mm
- Outer Diameter: 29 mm
- Length: 45 mm
- Max Speed: 4000 rpm
- Temperature Range: -40 to 350 deg Fahrenheit
- Set Screw: M5
- Seating Torque: 4 Nm

**11.** Mounted Ball Bearings with Two Bolt Flange - *Responsible for attaching shaft connected to the winch drums to IP-68 Electronic Enclosure*

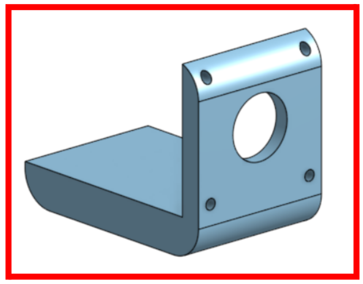
- Housing Material: Black-Oxide Steel
- Bearing Material: Chrome-Plated Steel
- Hole Mount Diameter: 5.5 mm
- For Shaft Diameter: 10 mm
- Seal Type: Sealed
- Height: 32 mm
- Width: 53 mm
- Length: 9.6 mm
- Flange Thickness: 2.2 mm
- Radial Load Capacity:
    - Dynamic: 380 lbs
    - Static: 175 lbs
- Max Speed: 18500 rpm
- Temperature Range: 0 - 210 deg Fahrenheit

**12.** Sealed Ball Bearing, 10mm - *Placed within the shaft end support to allow the shaft rotate within a fixed body*

- Outer Diameter: 26 mm
- Width: 8 mm
- Ring, Ball & Cage Material: Steel
- Seal Material: Buna-N Rubber
- Radial Load Capacity:
    - Dynamic: 1000 lbs
    - Static: 440 lbs

**13.** Motor Offset Stand

- Material: PLA
- Through Hole Diameter: 38.1 mm
- Motor Mount Diameter: 6.5 mm



**14.** 1045 Carbon Steel Rotary Shaft

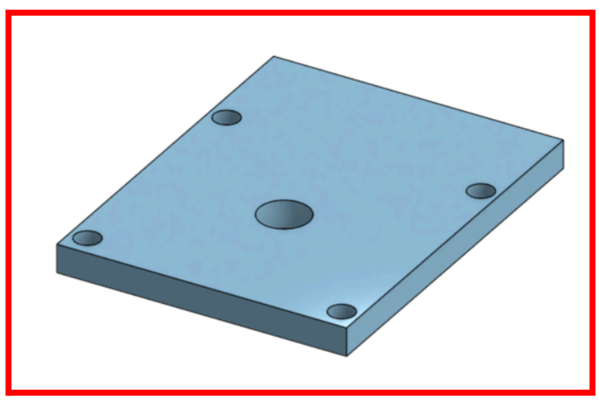
- Diameter: 10 mm
- Length: 100 mm
- Chamfered Edge
- Hardness: Brinell 197
- Mechanical Finish: Turned, Precision Ground, Polished

**15.** Bevel Gear. 20 tooth - *Used for Driven Gear of the Gear Train and allows for 1:2 gear reduction to help with increase in torque at lower speeds; an M3 was tapped into the hub of the gear to add a set screw*

- Face Width: 8 mm
- Module: 1.25
- Pressure Angle: 20 deg
- Gear Pitch Diameter: 25 mm
- Overall Width: 19 mm
- For Shaft Diameter: 8 mm
- Hub Diameter: 22 mm
- Hub Width: 10 mm

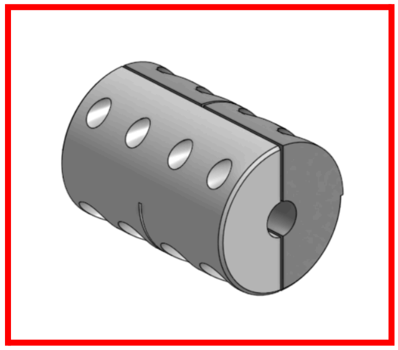
**16.** Top Bracket - *Used for gear alignment of drivergear in gear train and features four countersinks to connect M4 screws to heat set inserts of the Right and Left Bracket*

- Material: PLA
- Width: 82 mm
- Height: 106.3 mm
- Thickness: 8 mm
- Hole Diameter: 14.15 mm



**17.** Custom Two Piece Shaft Coupler - *Since there was no coupler to attach a 14 mm keyed motor shaft to an 8 mm shaft on the market, a customized 3d printed shaft with 100% infill was created; four M3 heat set inserts were inserted to constrain both of the shafts*

- Material: PLA
- Length: 50 mm
- Outer Diameter: 32.6 mm

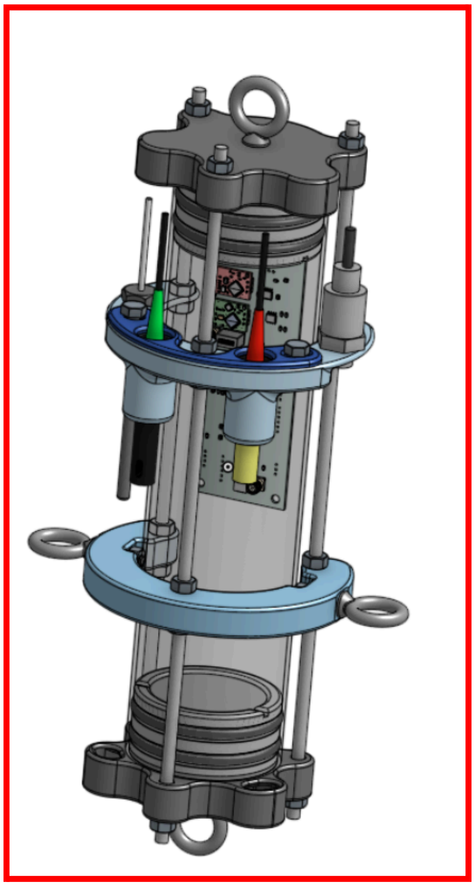


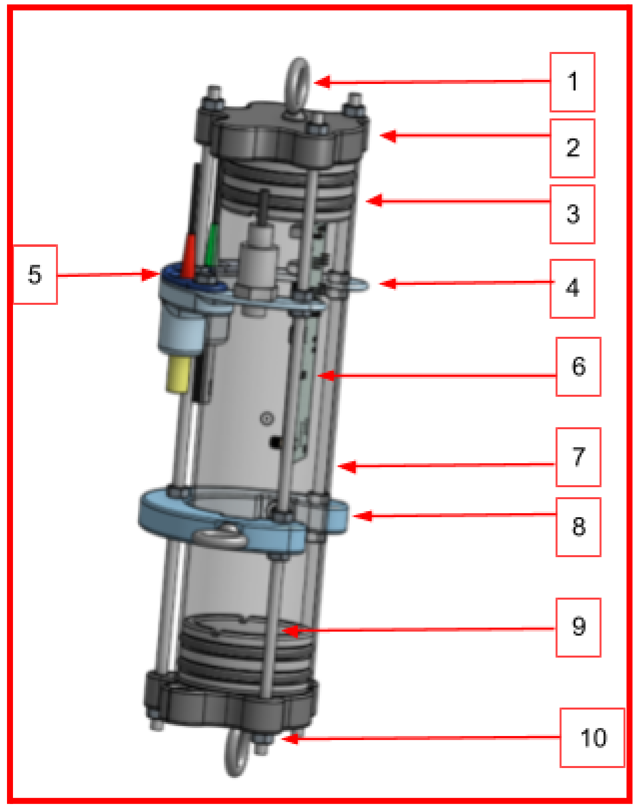
**18.** Keyed Rotary Shaft

- Material: 304 Stainless Steel
- Diameter: 10 mm
- Length: 400 mm
- Keyway:
    - Length: 400 mm
    - Width: 3 mm
    - Depth: 1.8 mm
- Hardness: Rockwell B90

generate

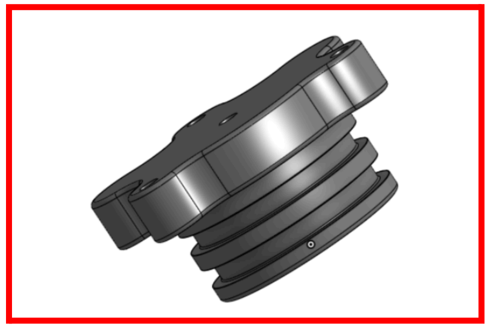
## Submersible System

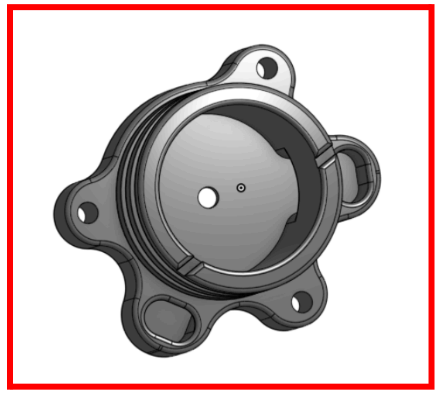





1. Steel Eye Bolt - Zinc-Plated, 1/4"-20 Thread Size, 1" Thread Length (4x)
    - Eye bolts were placed on the top, bottom and the sides of the submersible. The top hook was used to attach to the room connected to the winch system and the bottom hook was used to connect to the preexisting anchor on the ocean floor. The side hooks are used to attach to the kelp lines.
    - Material: Zinc-Plated Steel
    - Threaded size: 1/4"-20
    - Thread type: Fully Threaded
    - Height: 2 3/8"
    - Total cost: $65.00

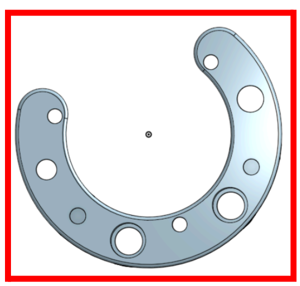
2. Submersible Cap (2x)
    - Encloses the submersible's body on both ends and has grooves to incorporate O-rings. Holes were created on the outside to allow rods to fit through. Two big holes were created from the insides to allow wires to pass through. After wires were placed, the holes were filled with epoxy to ensure the inside of the submersible would remain waterproof.
    - Material: PLA
    - Maximum height: 2.025 in
    - Maximum width: 3.492 in
    - Inside diameter: 2.319 in
    - Outside diameter: 2.920 in
    - Through hole size: 0.297 in

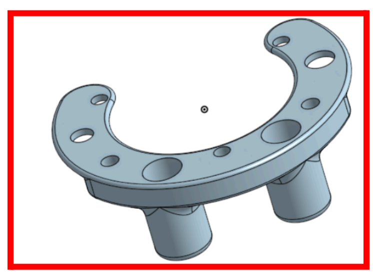




3. O-Ring (4x)
    - To ensure that the submersible was IP-69 and no water entered inside, O-rings were placed on the caps to make the system air tight.
    - Material: Nitrile Rubber
    - Inside diameter:  2-5/8"
    - Outside diameter: 3”
    - Width: 3/16"
    - Cost: $4.04

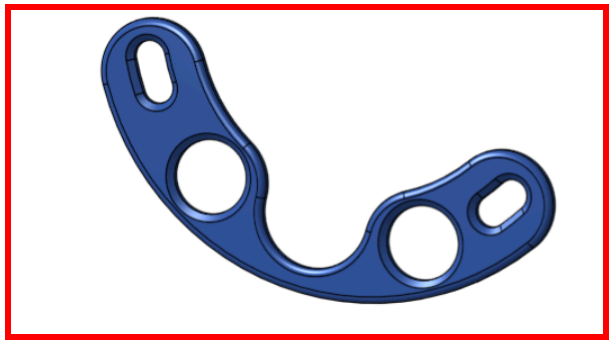
4. Sensor Bracket
    - Four sensors were used to measure various factors in the water and this bracket was created to hold them in place right outside of the submersible.
    - Material: PLA
    - Inside radius: 1.637 in
    - Outside radius: 2.575 in
    - Thread through hole: 0.297 in





- Sensor one through hole: 0.41 in
- Sensor two through hole: 0.55 in
- Sensor three through hole:  0.55 in
- Sensor four through hole: 0.5312 in

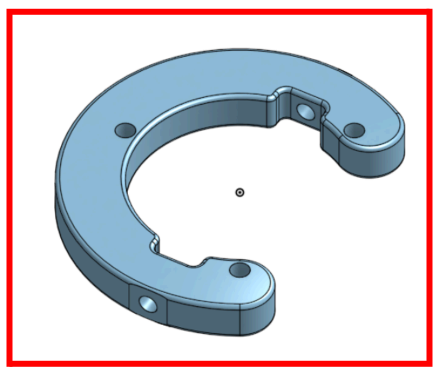
5. Sensor Bracket Cover
    - To stabilize the bigger sensors, a cover was added on top of the bracket.
    - Material: PLA
    - Inside radius: 1.637 in
    - Outside radius: 2.575 in
    - Thickness: 0.125 in



6. Submersible PCB
    - The PCB was stored inside the submersible and had the UV sensor attached to it.
    - Length: 5.384 in
    - Maximum thickness: 0.773 in
    - Width: 2.00 in

7. Submersible Body
    - The enclosure was created by using an off the shelf acrylic pipe.
    - Material: acrylic
    - Height: 6.00 in
    - Inside diameter: 2.912 in
    - Outside diameter: 3.070 in

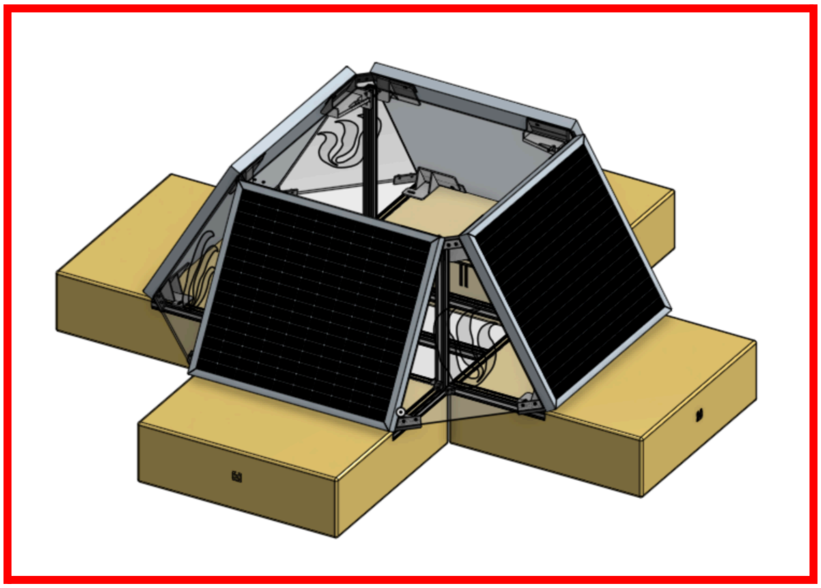
8. Submersible Hook Holder
    - To attach the side hooks, another bracket was created and was held in place by the rods.
    - Material: PLA
    - Inside radius: 1.637 in
    - Outside radius: 2.575 in
    - Thickness: 0.500 in

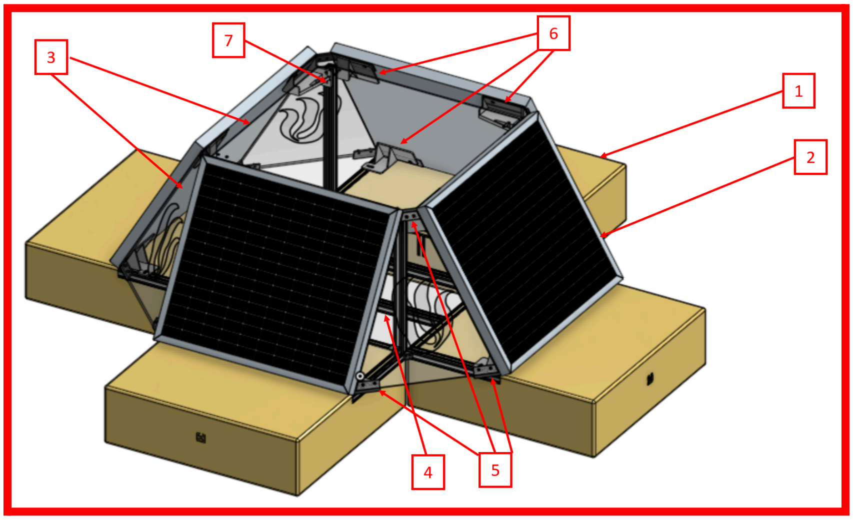


9. Threaded Rod - (3x)
    - To ensure all the brackets stayed in place, three rods running through the whole submersible were used.
    - Material: 316 Stainless Steel
    - Threaded size: 1/4"-20
    - Length: 1 ½ ft.
    - Cost: $15.33

10. Hex Nuts
    - Used to secure the rods
    - Material: 316 Stainless Steel
    - Threaded size: 1/4"-20
    - Thread type: UNC
    - Width: 7/16"
    - Height: 7/32"

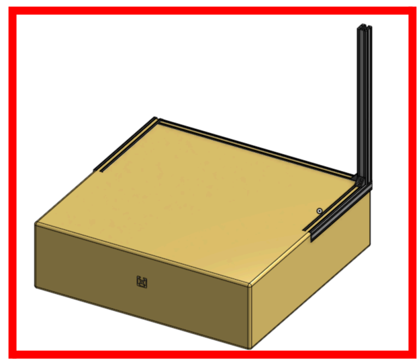
# Support Buoy System





1. Floating Block - *has an 80/20 interface to allow for mounting and is able to support approximately 600 lbs*

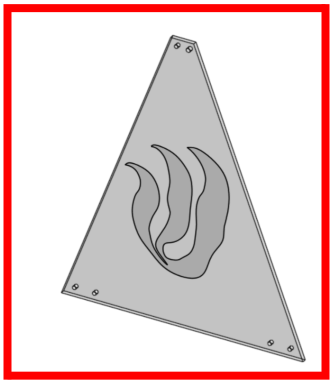
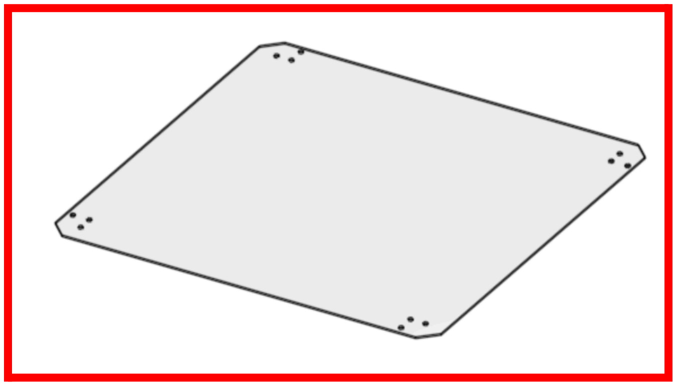
   - 80/20 Frame - *allows for hardware interaction with foam blocks without the damage that interfacing directly with the foam would cause*
     - Material: 1010 80/20 pieces
     - Length: qty x 2  11”
       qty x 1  20.9”
       qty x 1  2.43”
       qty x 1  21” rod
       qty x1  16”
     - Connectors: x7 80/20 corner brackets
     - Cost:
   - Foam block - *easily customizable with 150 lbs of support per gal*



     - Material: [TotalBoat Pourable Floation Foam]
     - Qty: 4 gal
     - 22.734” x 22” x 6.86” cube
     - Cost: $250
     - Amount of foam used could potentially be decreased. Due to non-ideal pouring conditions, maximum expansion was not achieved. Technically in perfect conditions 2 gals should be enough

2. Solar Panel - *Weather-proof with an easy to install frame already attached*
   - QTY: x 4
   - Cost: $74.99

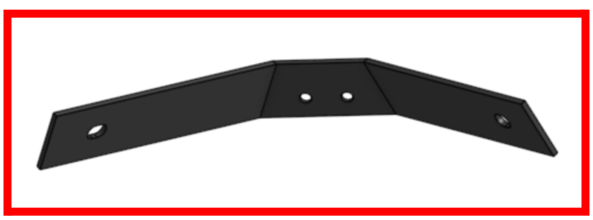
3. Panels - *Add some support to general structure will also providing some weather-proofing by blocking unfavorable weather conditions*
    - Material: Acrylic

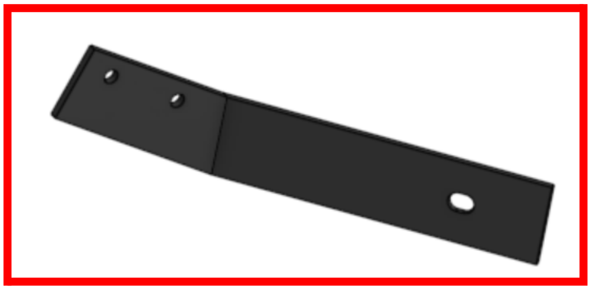
    - QTY: x4 side panels
      x1 top panel
    - Cost: $100

4. Winch Support Bars - *provide a base for the depth adjustment system*
    - Material: 1010 80/20
    - Length: 22.9”
    - QTY: 2
    - Cost:

5. Side Panel Mounting Brackets
    - Material: PLA
    - Top Panel Mount - *Provide mounting fixture for side panel and help connect two solar panels to each other*

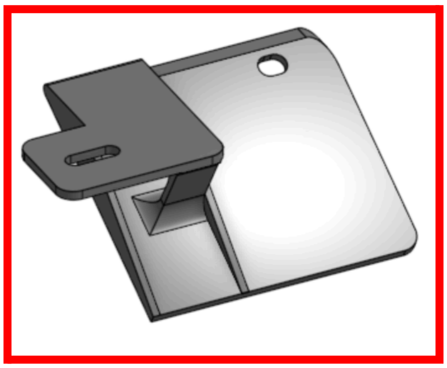


        - QTY: x4
    - Bottom Panel Mount (R & L) - *Provide mounting fixture for the side panels, left and right configurations are mirror images of each other*



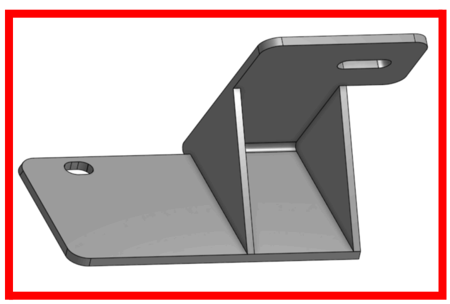
- QTY: x4 L Configuration
  x4 R Configuration

6. Solar Panel Mounting Brackets
   - Material: PLA
   - 30 Degree Bracket - *Connects the solar panel to the vertical 80/20 bar of its mated foam block*

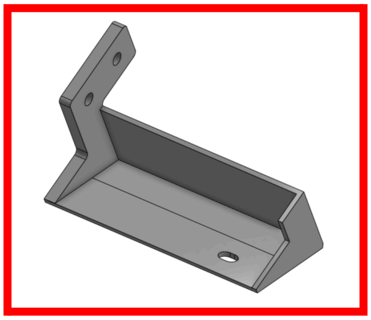


- QTY: x4

- 60 Degree Bracket (R & L) - *Connects the solar panel directly to the foam block (at 60 degrees since that is the optimal angle in Maine during the winter months)*
  - QTY: x4 L Configuration
  - QTY: x4 R Configuration

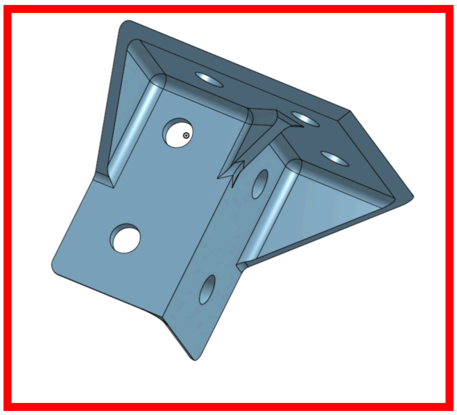


- Solar Bracket - *Connects the solar panel to the vertical 80/20 bar of its neighboring foam block*



    - QTY: x4

7. Top Panel Mounting Bracket - *Connects the top acrylic panel to the vertical 80/20 bars extruding from each floating block*



- Material: PLA
- QTY: x4

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# Electrical Design

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### Design Overview

- Flowchart of how components and subsystems interact
- Explanation of decisions made in using general technologies and concepts
- I/O of the electrical systems in relation to mechanical, software, etc.

The general electrical design can be separated into two main electrical subsystems: the buoy and the submersible.

**Buoy Electronics Overview:**
The buoy system handles the majority of electrical and software processes. It contains the data storage system, the motor driver board, and power converters (buck converter 50V-5V and LDO 5V-3V3). In the final system, an Arduino Uno was used as an external microcontroller. The power management system was located on the buoy and included a 50V battery, a solar charge controller, and 4 solar panels.

**Data Storage:**
To store collected data, information is saved on a 16GB microSD card located on the buoy. The buoy utilizes an ATMega328P as the main control unit. Since this microcontroller runs at 5V logic, signals between the ATMega and the SD card are shifted from 5V to 3V3 (and vice versa) using a level shifter. The current SD card circuit is designed for SPI communication.

In both the original and final data storage designs, the majority of components are implemented on the buoy PCB. There is one minor difference between the two. In the final version, the 3V3 power rail is sourced from the Arduino Uno's LDO instead of the board's LDO (although either configuration should work).

**Power Supply and Solar Charging:**
The entire power supply circuit is located on the buoy. The battery is a 50V rechargeable Lithium Ion battery designed for E-bikes and scooters. It runs the entire electrical system including the buoy PCB, the submersible PCB, and the motor.

Four 12V solar panels (50W) are arranged in series to provide a sufficient voltage to charge the battery. A solar charge controller handles charging the battery and ensures the rest of the electrical system receives sufficient power.

From the battery power rail, a buck converter outputs 5V. This 5V rail is used to power the electronics on the submersible and the majority of components on the buoy. A 3V3 power rail is obtained from the LDO on the Arduino Uno (for data storage circuit).

**Motor:**

The purpose of the motor was to connect to the gear drive train in order to wind and unwind the rope. Our original stepper motor was controlled with the Ponolulu stepper motor driver, which is placed on the buoy PCB. The motor should be able to draw up to 6A of current. The power source for the motor comes from this 48V battery. This source code for this motor can be found through the stepper motor library. If the BasicStepping.ino file is used, ensure that the current limit on line 52 is set to 4000 mA instead of 1000 mA. Unfortunately, this motor was not able to be programmed to draw enough torque to move the submersible up and down. Our motor controller also got fried before we could upload more code onto it.

Due to our motor controller not working a few days before Showcase, we had to switch to 12 V CIM motors.

**Submersible Electronics Overview:**

The submersible system focused solely on sensor integration. These include conductivity, temperature, UV light, pressure, and pH sensors (see the Sensors section for more specifics). The submersible also contained an Arduino Uno for processing sensor data and, subsequently, sending information to the buoy for storage.

**Communication and Cabling:**

The sole purpose of the submersible microcontroller is to process sensor data and then send information to the buoy. To achieve this, a cable containing four wires is connected from the buoy to the submersible. One of the wires provides a 5V power supply and another references Ground. The last two are utilized as UART communication lines so that the submersible can write sensor information to the buoy.

**Schematic and Layout**

The entire schematic and layout was designed in KiCAD 7.0. Footprints and symbols were obtained from Digikey, Mouser, SnapEDA, or designed ourselves. Each PCB was fabricated by JLBPCB. Screenshots of the rendered 3D CAD of each PCB can be found in the Appendix.

Both the submersible and buoy PCBs were designed for size and structure. The submersible PCB was designed to be longer so that it could fit in the tube-shaped submersible. The buoy PCB was designed to be as compact as possible.

The submersible PCB contained the required circuits/components for each sensor. The Atlas Scientific sensors (except for temperature) required electrical isolation circuits and spaces for their included circuits boards. The UV sensor required an op amp circuit to amplify its signal. The last two sensors, depth and temperature, only required SMA connectors. The original board also contained an ATMega328PB and an FTDI chip for USB programming/communication. Screw hole terminals were also included for emergency programming, UART communication with the buoy, and GPIO pins.

The buoy PCB contained the data storage circuit which included an SD card mount and a level shifter. It also provides space for the motor driver board. A buck converter and an LDO were also utilized to output power rails: 5V and 3V3. The original board also contained an ATMega328PB and an FTDI chip. Screw terminals were included for emergency programming, UART communication to the submersible, and GPIO pins.

### Sensors

- What is each sensor's purpose?
- Where should each sensor be placed?
- Include a link to the spec sheet and call out basic information that's relevant to the application like accuracy. Be sure to relate information to the application

**Conductivity Sensor -> Salinity**

- **Measurement Range:** 5 – 200,000 μS/cm (0.0032 -128 ppt)
- **Operable Range:**
  - **Temperature:** 1°C – 110°C
  - **Pressure/Depth:** Up to 500 PSI (1,157 ft depth)
- **Default I2C Address:** 100 (0 x 64)

The conductivity sensor is connected on the submersible PCB (see figure 2 in appendix) through the SMA connector connected to its corresponding conductivity circuit board. The Atlas-Scientific/Ezo_I2c_lib library is used to generate sensor readings of up to two decimal places. Salinity can be calculated using the practical salinity scale which utilizes conductivity and temperature to estimate salinity levels.

[pH Sensor](#)

- **Measurement Range:** 0 - 14 pH
- **Operable Range:**
    - **Temperature:** -5°C – 99 °C
    - **Pressure/Depth:** Up to 110 PSI (254 ft depth)
- **Default I2C Address:** 99 (0 x 63)

The pH sensor is connected on the submersible PCB (see figure 2 in appendix) through the SMA connector connected to its corresponding pH circuit board. The Atlas-Scientific/Ezo_I2c_lib library is used to generate sensor readings of up to two decimal places. Kelp is best grown with pH values of 7.0-9.0. These readings are not expected to change as much because the natural ocean range remains relatively constant.

Temperature Sensor

- **Measurement Range:** -50 - 200 °C
    - Expected Range: 2 - 20°C
- **Operable Range:**
    - **Temperature:** -50°C - 200°C
    - **Pressure/Depth:** Not Specified

The temperature sensor can be connected to the rightmost SMA connector on the submersible PCB. The optimal conditions for kelp is around 41F-59F (5C-15C).

Depth/Pressure Sensor

- **Measurement Range:** 0 - 290.08 PSI
    - Expected Range: 15.1 - 41.3 PSI
- **Operable Range:**
    - **Temperature:** -20°C ~ 85°C
    - **Pressure/Depth:** Up to 290.08 PSI (635 ft depth)

The depth sensor contains a 3 pin surface which includes VCC, GND, and an analog input connected to the submersible PCB. The purpose of this sensor is to gauge ocean depth, and whether it is in the optimal position for kelp growth.

UV Light Sensor

- **Measurement Range:** 240 ~ 370 nm
    - Expected Range: 290 ~ 400 nm
- **Operable Range:**

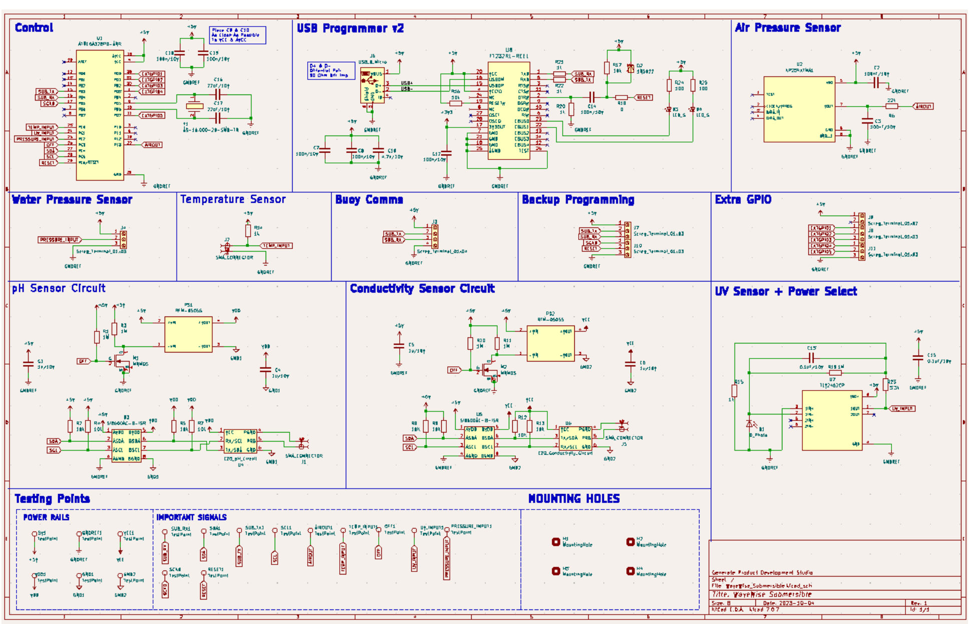
**Temperature:** -30°C ~ 85°C

The UV sensor is connected with an op amp circuit on the submersible PCB.

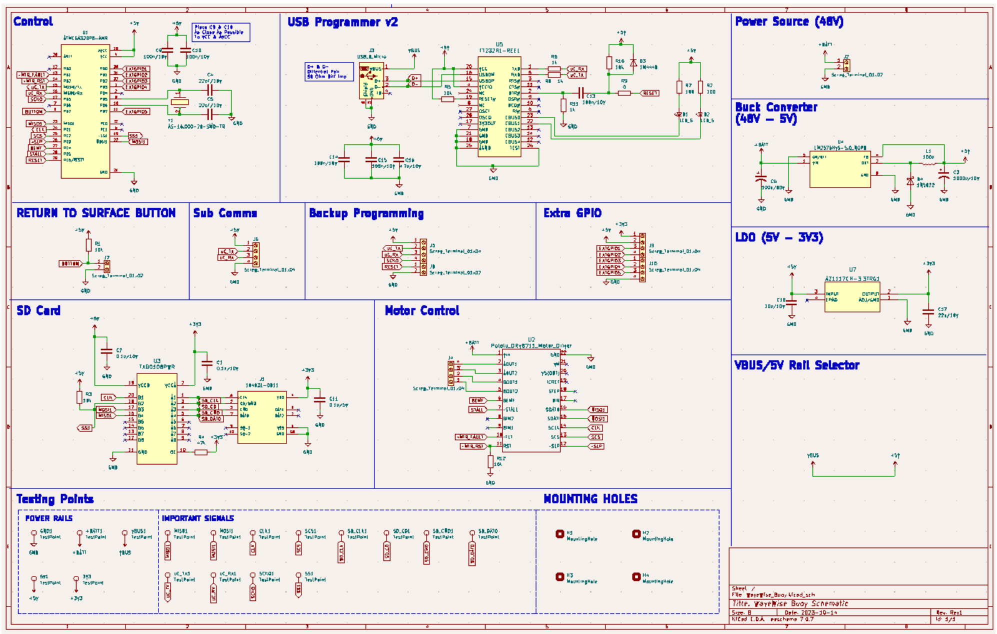
## Circuit Board

- Schematics
- Any input / output and where it connects to sensors, mechanical devices, etc.
- Images to show relative size and placement within the overall system

**Submersible Schematic:**



## Buoy Schematic



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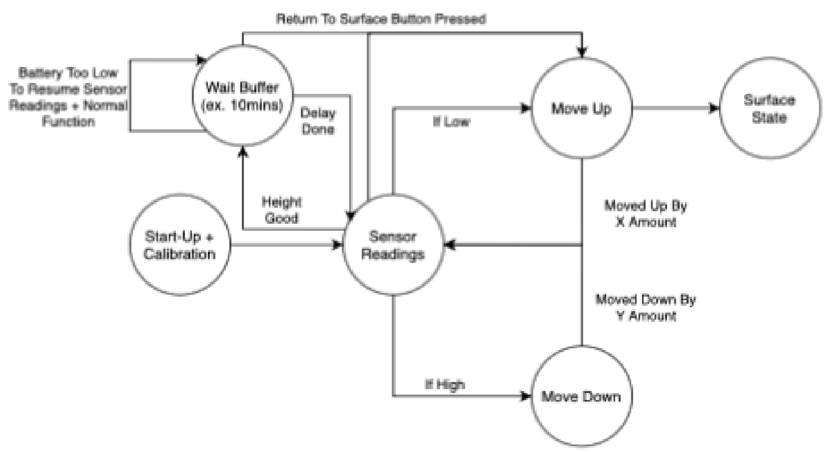
# Software Design

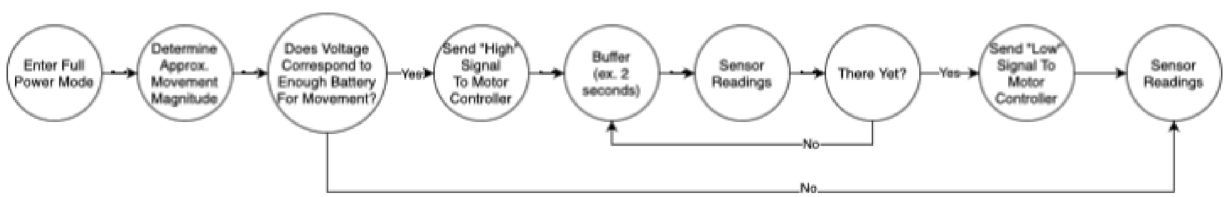
## General Software Design

- Development Environment
- Required libraries and tools
- Flowchart of software flow

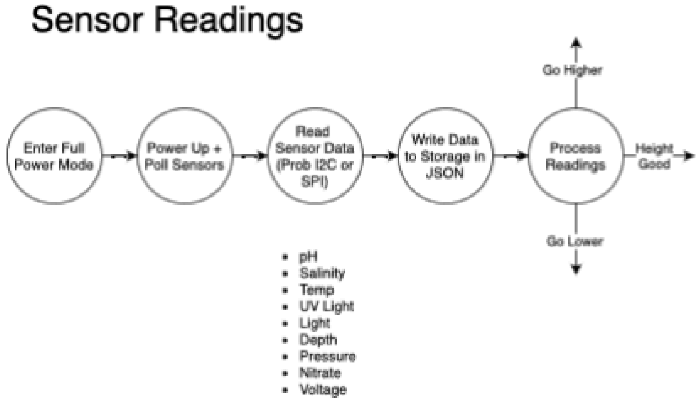
### Overall




### Move Up/Down




### Sensor Readings

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All code for both the submersible and buoy software was written in C++, using the Arduino IDE and Visual Studio Code. The only downloaded library was the EZO_I2C header file that could be found from the Atlas Scientific GitHub. The other added libraries were the SPI, SD, SoftwareSerial, Servo, Wire, and Sequencer2 headers (these can be seen in the Appendix below).

## Explanation of Overall Software Decisions

**Buoy:**

The main component of the buoy code is the SD card. Data readings were transmitted from the submersible to the buoy, and printed out on a new .txt file. The final prototype utilizes Arduino's SD card library to open and write information to the SD card. No information is ever read from memory (although it would be easy to implement with the Arduino library). In the current version, information is received from a software declared UART bus (located on digital pins). Data is received as characters and blocks of information are separated by newline characters.

Motor control is the only other main software feature of the buoy. The final prototype utilizes a CIM DC motor which cannot reliably move certain distances. Because of this, the motor is programmed to move up and down at short intervals to test functionality. There is no code that currently ties sensor input to motor control. However, both subsystems are fully functional and should, in theory, be able to communicate with each other to determine distance.

**Submersible:**

The submersible software focused solely on processing sensor data and transmitting information via UART to the buoy. This means the submersible code is straightforward and simplistic. Information is received from the sensors every 10 seconds. This raw data is then converted into relevant metrics (such as determining temperature based on voltage input). Then, using Arduino's SoftwareSerial library, information is sent via a software imitated UART bus to the buoy. Sensor readings are transmitted in the form of characters instead of any binary or floating point equivalent to the decimal calculations.

**Communication**

- Command set
  - Commands available
  - How to interpret
- Settings
  - Baud rate, necessary connections, frequency, etc.

# Next Steps

- Define exactly where the project stands
  - What works? What doesn't?
  - What are the existing projects / shortcomings?
- What do you recommend doing next?
  - Getting quotes from manufacturers?
  - Gathering user feedback
  - What would you do if you had more time to work on the project?

### Software Design

- Currently, the motor is able to move the submersible up and down and all of the sensors are able to produce readings. However, code needs to be written in order to control how the sensor readings affect the motor movement for optimal depth.
- The sensors are able to produce readings, but more testing needs to be done in order to determine their accuracy. If calibration needs to be done, a proposed solution is Kalman filtering. Some initial research can be found here:
  - Kalman Filter Initial Research 1
  - Kalman Filter Initial Research 2

# Works Cited

[1] Matthew Bennion, Jessica Fisher, Chris Yesson & Juliet Brodie (2019) Remote Sensing of Kelp (Laminariales, Ochrophyta): Monitoring Tools and Implications for Wild Harvesting, Reviews in Fisheries Science & Aquaculture, 27:2, 127-141, DOI: 10.1080/23308249.2018.1509056

[2] Flavin, K. & Flavin, Nicholas & Flahive, B.. (2013). Kelp Farming Manual a Guide to the Processes, Techniques, and Equipment for Farming Kelp in New England Waters. Kelp farming manual. A guide to the processes, techniques and equipment for farming kelp in New England Waters.

[3] J. N. Snyder, T. W. Bell, D. A. Siegel, N. J. Nidzieko, and K. C. Cavanaugh, "Sea surface temperature imagery elucidates spatiotemporal nutrient patterns for offshore kelp aquaculture siting in the Southern California bight," Frontiers in Marine Science, vol. 7, 2020. doi:10.3389/fmars.2020.00022

[4] D. Sigman, "The Biological Productivity of the Ocean | Learn Science at Scitable," www.nature.com, 2012. https://www.nature.com/scitable/knowledge/library/the-biological-productivity-of-the-ocean-70631104/

[5] "Ocean Conditions | U.S. Climate Resilience Toolkit," toolkit.climate.gov. https://toolkit.climate.gov/topics/marine/ocean-conditions

[6] "Effects of depth-cycling on nutrient uptake and biomass production in the giant kelp Macrocystis pyrifera," Renewable and Sustainable Energy Reviews, vol. 141, p. 110747, May 2021, doi: https://doi.org/10.1016/j.rser.2021.110747.

[7] "Eastport, Maine Travel Weather Averages (Weatherbase)," Weatherbase. https://www.weatherbase.com/weather/weather.php3?s=80627&cityname=Eastport-Maine-United-States-of-America&units=us (accessed Dec. 19, 2023).

[8] “How Cold Weather Affects Electronics,” Global Electronic Services, Dec. 11, 2017. https://gesrepair.com/how-cold-weather-affects-electronics/

[9] “How Does Cold Weather Affect Your Car Lubricants?,” Petroleum Service Company. https://petroleumservicecompany.com/blog/how-cold-weather-affects-car-lubricants/

# Appendix

## PCB Designs

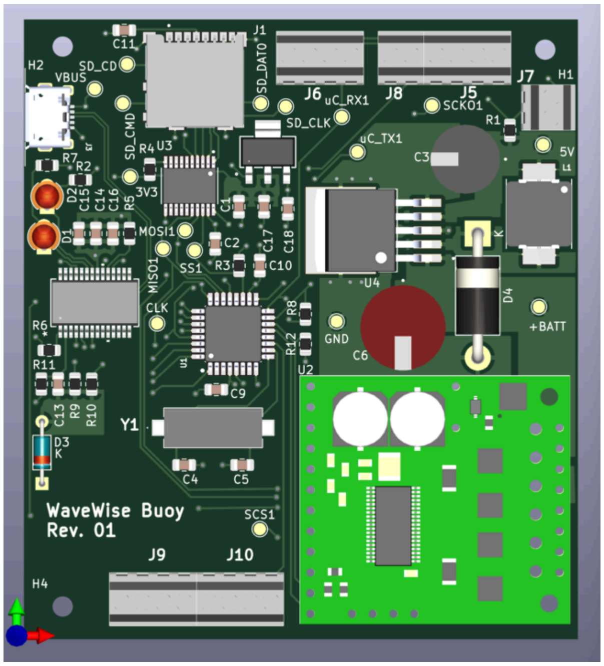


Figure 1. Buoy PCB

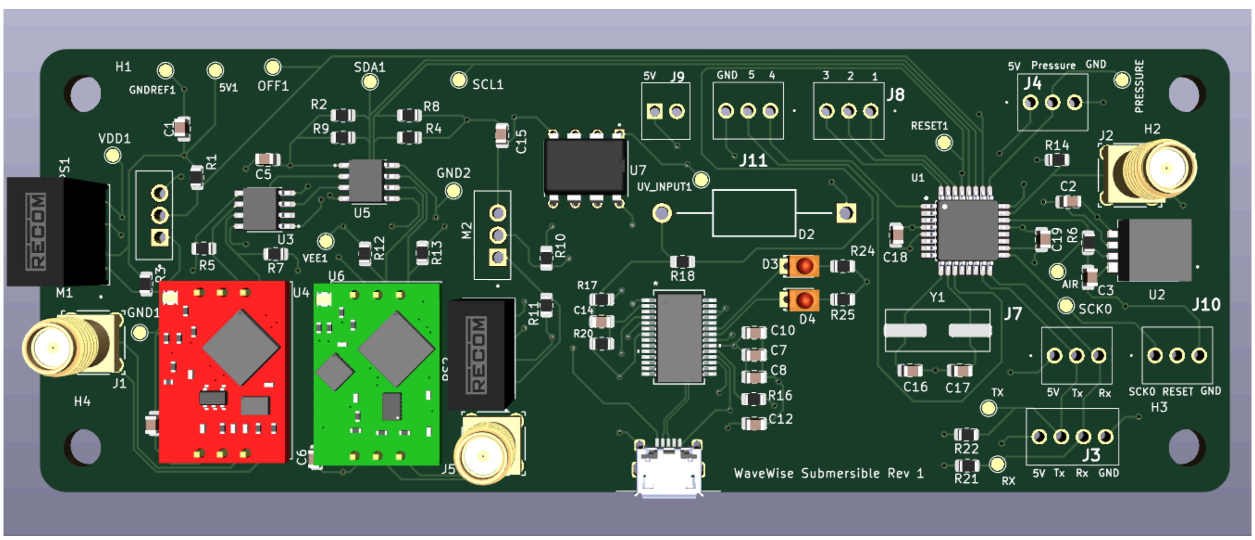


Figure 2. Submersible PCB

## General Code

Submersible: Sub.ino

Buoy: Buoy.ino